BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI

II SEMESTER 2013-2014

EEE/CS/INSTR F241 MICROPROCESSOR PROGRAMMING AND INTERFACING

COMPREHENSIVE EXAMINATION-A (OPEN BOOK)

MARKS: 30 **13-05-2014** **DURATION: 60 MIN**

ID No:	**Name:**	**Sec:**

Note: This in-built question/answer paper contains fifteen questions, each carrying 2 marks. Use last few pages of answer sheet of PART-B to do your rough work. You can submit PART-A anytime and collect PART-B question paper.

Recheck Request:

Q1. The expression ***No. of clocks = 22(CX-1) + 1,441,802 (DX-1) + 36*** introduces a delay of 0.758 sec for CX=4150H and DX=0006H in an x86 CPU based program. The frequency of operation of CPU is 10 MHz. Compute the minimum and maximum delay which could possibly be achieved using the expression.

Q2. A MACRO is written by a user to push the contents of EAX, EBX, ECX and EDX registers in order onto the stack. Show the contents using register names and their physical addresses in the stack, after the code associated with the MACRO is executed. Initially, assume SS=0000H and SP=1000H.

Q3. If a media has 256 bytes/sector and the maximum number of root directory entries specified in BIOS Parameter Block is 612, then total number of sectors reserved for root directory is:

Q4. If the 8 byte descriptor of a segment in 80286 is 00 00 FF 32 00 00 00 FF, then the size and the starting address of the segment is

Q5. If AX=FFFFH and CL = 02H. What is the result of executing the instruction DIV CL?

Q6. The USB protocol uses NRZI encoding scheme to encode the data before transmitting it. Use this scheme to encode the following bits: 111101101

Q7. Out of the three inputs of 8086, NMI, INTR and HOLD, which input has the highest priority?

Q8. Compute the time taken to transmit three ASCII characters via 16550 operating at 19200 baud rate with 8 bit data, odd parity and one stop bit.

Q9. For 80486 CPU, what are the four different methods using which an application running at the lowest privilege level can access the procedure running at the highest privilege level?

Q10. For 80386, answer **True/False** for the following statements:

a) The CPL of the currently executing procedure is normally equal to the privilege level of the code segment from which the instructions are being fetched.

b) A CALL gate with a DPL of 2 can be used by a program with a CPL of 2 to access a system service at level 0 but the same gate cannot be used by a program with a CPL of 3.

Q11. (a) A cache is divided into 16 sets of 4 lines each. The main memory contains 4K blocks of 128 words each. The format of the main memory addresses would be

TAG:	SET:	WORD/OFFSET:

(b) Conflict miss is higher in direct mapped cache as compared to set associative cache (True/False)

Q12. A computer has a cache, main memory and a hard disk. If an operand is in the cache, 20 ns are required to access it. If it is in the main memory but not in cache, 60 ns are needed to load it into cache and then it is accessed by CPU. If the operand is not in main memory, 12 ms are required to fetch it from disk, followed by 60 ns to copy it into the cache and then the operand is accessed by the CPU. The cache hit ratio is 0.9 and the main memory hit ratio is 0.6. What is the average time in nano seconds required to access the required operand on this system?

Q13. .The Intel 8088 consists of a bus interface unit (BIU) and an execution unit (EU), which forms a 2-stage pipeline. The BIU fetches instructions into a 4-byte instruction queue.

a) Suppose the tasks performed by the BIU and EU take about equal time. By what factor does pipelining improve the performance of the 8088? Ignore the effect of branch instructions.

b) Repeat the calculation assuming that the EU takes twice as long as the BIU.

Q14. Consider a system employing interrupt-driven I/O for a particular device that transfers data at an average of 8 KB/s on a continuous basis. Assume that interrupt processing takes about 100 s (i.e., the time to jump to the interrupt service routine (ISR), execute it, and return to the main program). Determine what fraction of processor time is consumed by this I/O device if it interrupts for every byte.

Q15. A pipelined processor has a clock rate of 2.5 GHz and executes a program with 1.5 million instructions. The pipeline has five stages, and instructions are issued at a rate of one per clock cycle. What is the throughput (in MIPS) of the pipelined processor?

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